

Average Payoff, Typical Ticket

AP Statistics · Unit 2 · Topic 2.8 · B: 2026-10-30 · E: 2026-11-20 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU VAR-5** — Probability distributions may be used to model variation in populations.
- **LO VAR-5.A** — Represent the probability distribution for a discrete random variable. [Skill 2.B]
- **EK VAR-5.A.1** — The values of a random variable are the numerical outcomes of random behavior.
- **EK VAR-5.A.2** — A discrete random variable is a variable that can only take a countable number of values. Each value has a probability associated with it. The sum of the probabilities over all of the possible values must be 1.
- **EK VAR-5.A.3** — A probability distribution can be represented as a graph, table, or function showing the probabilities associated with values of a random variable.
- **EK VAR-5.A.4** — A cumulative probability distribution can be represented as a table or function showing the probability of being less than or equal to each value of the random variable.
- **LO VAR-5.B** — Interpret a probability distribution. [Skill 4.B]
- **EK VAR-5.B.1** — An interpretation of a probability distribution provides information about the shape, center, and spread of a population and allows one to make conclusions about the population of interest.

Essential question: How can expected value and the most likely outcome both be true while answering different questions?

DOK-3 skill: 4.B · **FRQ pattern:** expected-value-vs-typical-outcome-what-the-mean-tells-one-person

DOK rationale: Part (c) coordinates a probability distribution, gross and net expectations, and the modal outcome to reconcile two true summaries, assess a value claim, and defend an individual decision.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask for a buy-or-decline commitment without calling the raffle fair or unfair.
- Begin the combined 4.7–4.8 video follow-along at minute 5.
- Return to the distribution at minute 33. Circulate and separate gross payout, net profit, expected value, and modal outcome.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Make a first decision using one number or possible outcome.
- Complete the random-variable and probability-distribution portion of the follow-along.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What is subtracted from every prize when X is net profit?
- How can a few rare large values move a mean while most outcomes remain zero gross payout?
- Is negative 50 cents one of the possible outcomes for a single ticket?
- Are you evaluating the ticket only as an investment, or does supporting the club have value in your criterion?

Adult role

Solo teacher. Circulate 5 pairs. Require students to reconcile both claims before evaluating the deal; allow different buying decisions only when the criterion and expected loss are both acknowledged.

[WARN] WATCH FOR

- Using gross payout as net profit and forgetting the two-dollar cost.
- Taking an unweighted average of the four possible values.
- Treating expected value as either the most likely value or a guaranteed single-ticket result.

SCORING PART (c) — E / P / I

Expected elements

- **reconciles-mean-and-mode** (required) — Explains how a 1.50-dollar mean gross payout can coexist with an 87.5 percent chance of no prize
- **uses-net-expectation** (required) — Subtracts the ticket cost and uses the negative 0.50-dollar expected net to assess the money-making claim
- **bounds-one-buyer-meaning** (required) — Interprets expected value as a long-run average and states that it does not predict an individual ticket's outcome
- **defends-buying-decision** (required) — Makes a decision tied to an explicit financial or nonfinancial criterion rather than treating expected value as a command

E	Reconciles both true summaries, rejects the financial good-deal claim with the net expected value, correctly bounds what the mean says about one ticket, and defends a decision with a stated criterion.
P	Computes or uses the negative net expectation and recognizes that most tickets lose, but omits the reconciliation, individual-outcome limit, or criterion behind the decision.
I	Calls the raffle profitable because the gross mean is positive, says an individual ticket will lose exactly 50 cents, or claims most losing tickets make a 1.50-dollar mean impossible.

Common mistakes

- Forgetting to subtract the \$2 ticket cost from each prize
- Averaging the four possible dollar values without probability weights
- Treating \$1.50 as the most common payout
- Interpreting an expected loss of \$0.50 as a guaranteed outcome for every buyer

[STAR] TODAY'S PROBLEM — annotated

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	-2	175/200

FIRST TAKE (5 min)

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

Key: Not graded. Students may focus on the top prize or on the 175 losing tickets; the distribution should connect those views.

Rules callout “DISTRIBUTION, EXPECTATION, AND ONE OUTCOME (keep on the page)” is printed on the student sheet.

DOK 1 (a) State the four possible values of X and identify the most likely value.

Key: The possible net profits are 98, 23, 3, and -2 dollars. The most likely value is $X = -2$, because 175 of the 200 tickets pay no prize.

DOK 2 (b) Verify that the probabilities form a distribution. Calculate the expected gross payout and the expected net profit μ_X .

Key: The probabilities sum to $1/200 + 4/200 + 20/200 + 175/200 = 200/200 = 1$. Expected gross payout is $100(1/200) + 25(4/200) + 5(20/200) + 0(175/200) = 1.50$ dollars. Expected net profit is $\mu_X = 98(1/200) + 23(4/200) + 3(20/200) - 2(175/200) = -0.50$ dollar per ticket. Equivalently, $1.50 - 2.00 = -0.50$.

DOK 3 (c) Decide whether Priya's numerical statement and Mateo's statement can both be true, and explain why there is no contradiction. Assess Priya's good-deal conclusion using the ticket cost and μ_X . Explain what expected value does and does not predict for one buyer, then decide whether you would buy and defend the criterion behind your choice.

Key: Both numerical statements are true. The mean gross payout is 1.50 dollars even though $175/200 = 87.5\%$ of tickets pay no prize, because the few large prizes pull the mean above the typical payout of zero. Priya's good-deal conclusion is not supported as a money-making claim: after the \$2 cost, the long-run average net is a loss of 0.50 dollar per ticket. Expected value predicts the average net over many repetitions under this distribution; it does not say one buyer will lose exactly 0.50 dollar. One ticket yields only 98, 23, 3, or -2 dollars. A student may reasonably decline on financial grounds, or buy if supporting the club and entertainment are worth more than the expected loss, provided that nonfinancial criterion is explicit.

EXIT

One thing the video changed about my first take: _____